

RAFFLES INSTITUTION
2013 Year 6 Preliminary Examination
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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BIOLOGY

Paper 3

9648/03

23 September 2013

2 hours

Additional materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, **hand in your essay SEPARATELY**.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/13
2	/12
3	/15
4	/12
5	/20
Total	/72

This document consists of 16 printed pages.



Raffles Institution
Internal Examination

Answer **all** questions.

- 1 (a)** Restriction mapping is a technique used to identify restriction enzyme cleavage sites. In an experiment, restriction mapping was used to investigate an unknown piece of DNA. This piece of DNA was 3 000 base pairs (bp) long.

The scientists first took plasmids that had one restriction site for the enzyme *KpnI* and one restriction site for the enzyme *BamHI*. The unknown piece of DNA was ligated into the multiple cloning site of the plasmid.

Fig. 1.1 shows the recombinant plasmid.

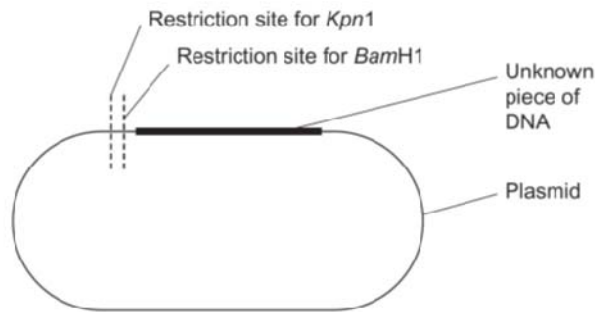


Fig. 1.1

When the scientists digested one of the recombinant plasmids with *KpnI*, they obtained two fragments. One fragment was measured as 1000 bp. The other fragment was described as "very large".

- i. What does this show about the unknown piece of DNA?

.....

 [2]

- ii. One of the fragments that the scientists obtained was described as "very large". What is represented by this very large fragment?

..... [1]

- iii. When the scientists digested another of the recombinant plasmids with *Bam*HI, they obtained three fragments. How many *Bam*HI restriction sites are there in the unknown piece of DNA?

..... [1]

- iv. When carrying out restriction mapping of the unknown DNA, scientists need to make sure that the restriction enzyme they have used has completely digested the DNA.

Suggest how they could use results from agarose gel electrophoresis to show that this DNA has **not** been completely digested.

.....

..... [1]

- (b) The urate transporter is a crucial protein found primarily in the kidney. **Fig. 1.2** below shows how genetic engineering can be used to clone the human urate transporter gene in bacteria.

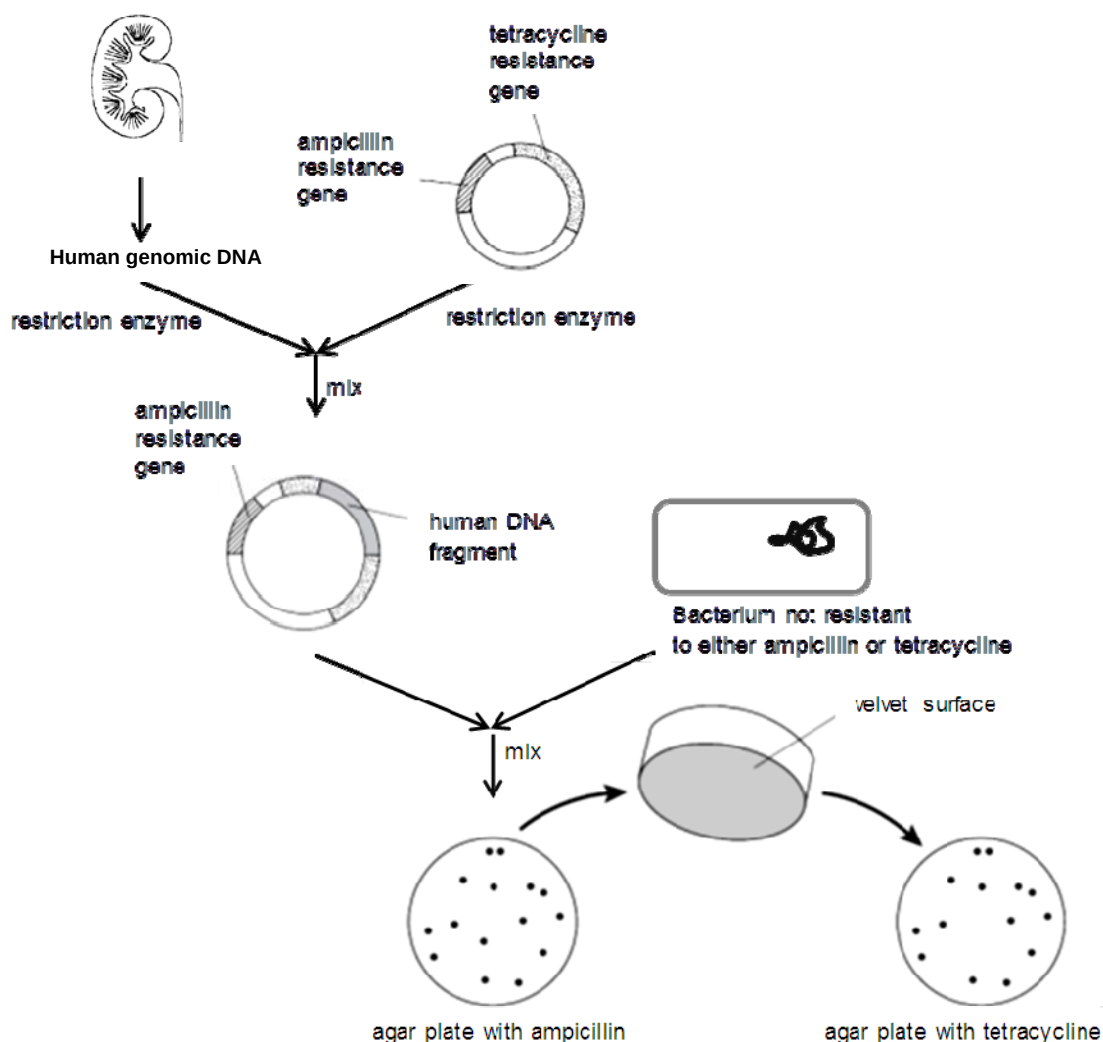


Fig. 1.2

- i. The human genomic DNA sample was digested with restriction enzyme. What is an important assumption made in this procedure to clone the urate transporter gene?

.....
 [1]

- ii. Comment on the choice of using the kidney for the extraction of the urate transporter gene in the procedure shown in **Fig. 1.2**.

.....
.....
.....
..... [2]

- iii. The human DNA fragments are inserted into the gene coding for resistance to tetracycline. What effect will this have?

.....
.....
.....
..... [2]

- iv. During replica plating, why is the velvet surface used?

.....
..... [1]

- v. Use an arrow (→) to point to the bacterial colony containing the human DNA fragment. [1]

- vi. State one disadvantage of using the bacterium as a host cell for expressing the urate transporter gene.

.....
..... [1]

[Total: 13]

- 2 Primordial dwarfism (PD) is a rare autosomal recessive disorder in which the lack of normal growth is not due to a deficiency of growth hormone. In linkage analysis using RFLPs, scientists can examine segments of DNA for a haplotype that is associated with PD. A haplotype refers to a group of gene loci or genetic markers that is inherited together on one chromosome. An individual inherits one haplotype from each parent.

Fig. 2.1 shows a pedigree of a family affected by PD followed by haplotype analysis with polymorphic microsatellite markers on a segment of chromosome 3. Each microsatellite marker has 7 different known alleles.

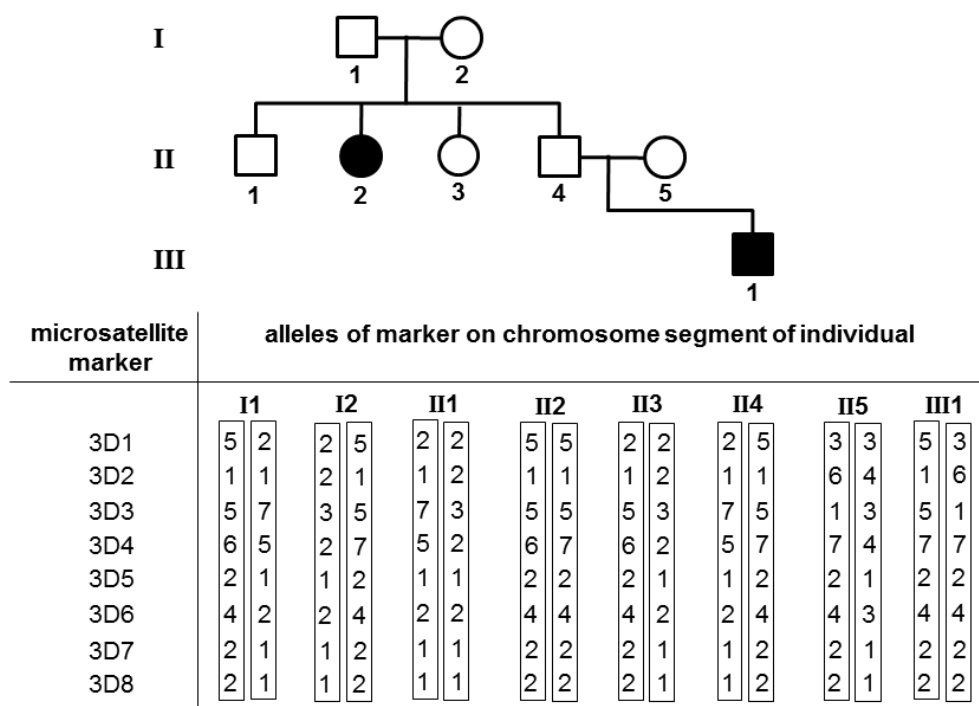


Fig. 2.1

- (a) By examining the haplotypes in **Fig. 2.1**,
- identify all the microsatellite marker alleles that are associated with primordial dwarfism.
.....
..... [2]
 - identify which descendant is a recombinant.
..... [1]

- (b)i.** One of the conditions for a microsatellite marker to be useful for disease detection is that it must be polymorphic.

Explain why the following 2 conditions are also necessary:

Condition 1: locus of the microsatellite marker allele must be tightly linked to the gene locus of the disease allele;

.....
..... [1]

Condition 2: restriction sites must flank the microsatellite marker allele.

.....
..... [1]

- ii.** Suggest why the disease allele causing PD can only be mapped using linkage analysis.

.....
..... [1]

After more research, scientists discovered another RFLP marker that is also linked to PD. **Fig. 2.2** shows the two alleles of the RFLP marker with their restriction sites.

They carried out PCR to amplify the target DNA segment containing the RFLP allele. The amplified DNA can then be subjected to PCR-RFLP analysis where restriction digestion using *BtgI* is done, followed by gel electrophoresis. The products of restriction digestion using *BtgI* can also be subjected to Southern blot analysis.

Fig. 2.3 shows the results of the PCR-RFLP analysis and the Southern blot analysis.

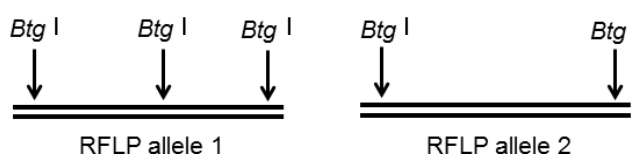


Fig. 2.2

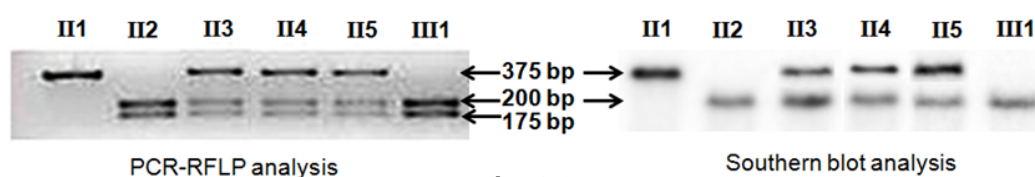


Fig. 2.3

(c) Outline the principle of gel electrophoresis.

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.....

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..... [3]

(d) Using information from **Fig. 2.2** and **Fig. 2.3**,

- indicate on the RFLP allele 1 in **Fig. 2.2** the position where the probe should hybridise; [1]
- explain the banding pattern of individual **II4** as seen in the PCR-RFLP analysis in **Fig. 2.3**.

.....

.....

.....

..... [2]

[Total: 12]

- 3 The *Syngonanthus mucugensis* species has significant economic value for the Brazilian ornamental dry flower business. However, extreme exploitation has rendered this species endangered.

Attempts at conventional propagation of this species by seeding have generated little success and studies are ongoing to develop alternative techniques that can assure propagation on a commercial scale. In a particular experiment, the potential of three different explants (stem, leaf and root) to form callus was studied using two plant growth regulators (BAP and NAA). The data from the experiment is shown in **Table 3.1**.

Table 3.1

PGR (μM)		% of callus		
BAP	NAA	Stem	Leaf	Root
0	0	0	0	0
	1.34	0	0	0
	2.68	8.62	19.73	0
2.22	0	0	0	0
	1.34	0	0	0
	2.68	0	0	0
4.44	0	0	12.35	0
	1.34	0	0	0
	2.68	0	0	0

- (a) With reference to **Table 3.1**,

- i. identify which explant seemed most suitable for use in micropropagation.

..... [1]

- ii. explain your answer to (a) i.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) In a separate experiment, the ability of the explant to undergo organogenesis was studied using the same plant growth regulators. Data obtained is shown in **Table 3.2**.

Table 3.2

PGR (μM)		% of shoots		
BAP	NAA	Stem	Leaf	Root
0	0	0	0	0
	1.34	0	0	0
	2.68	0	0	0
2.22	0	32.18	0	0
	1.34	0	0	0
	2.68	0	0	0
4.44	0	47.05	0	0
	1.34	0	0	0
	2.68	0	0	0

With reference to **Table 3.2**, explain which plant growth regulator has an inhibitory effect on shoot regeneration.

.....

.....

.....

.....

.....

..... [3]

To create transgenic maize which can produce the *Bacillus thuringiensis* toxin, the recombinant plasmid shown in **Fig. 3.1** was constructed. The recombinant plasmid also contains the gene for phosphomannose isomerase (PMI), an enzyme which allows plants cells to utilise mannose as a carbon source.

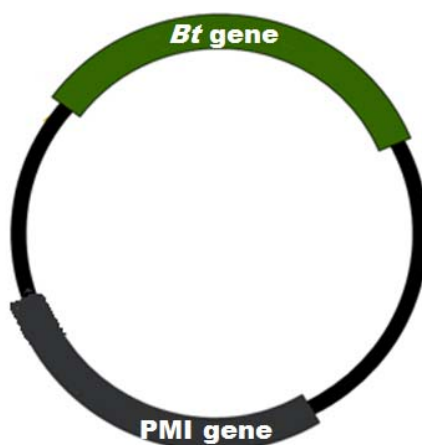


Fig. 3.1

To produce transgenic maize using the plasmid shown in **Fig. 3.1**, the procedure shown in **Fig. 3.2** was carried out.

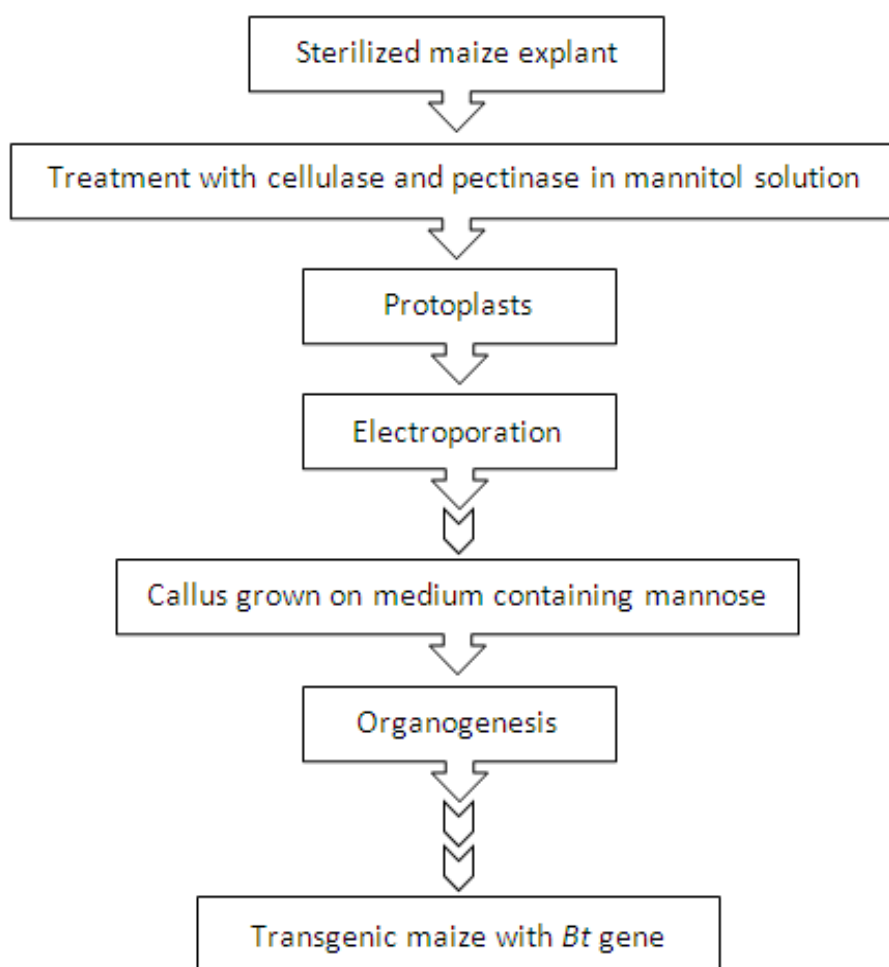


Fig. 3.2

(c) With reference to **Fig. 3.2**,

- i. explain why the sterilised explant was treated with cellulase and pectinase before electroporation.

.....

.....

.....

..... [2]

- ii. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplast is formed even if the sterilised explant was treated with cellulase and pectinase.

State what happens to the protoplasts if mannitol was not used.

..... [1]

- (d) Antibiotic or herbicide resistance genes are commonly included in the plasmid used for creating transgenic plants.

Explain why such genes were not needed in this case.

.....
.....
.....
.....[2]

- (e) Explain the benefit of incorporating a *Bt* gene into a crop plant.

.....
.....
.....
.....[2]

[Total: 15]

Planning question

- 4 The purple colour of beet root (*Beta vulgaris*) is due to the water soluble betacyanin pigment molecules that are found in the vacuole of intact cells. Beet root pigment is used commercially as a food dye and heat is typically used as a method to extract the pigment from intact beet root cells.

A colorimeter can be used to measure the absorbance of light of 550nm by betacyanin solution. The concentration of betacyanin is proportional to its absorbance. The colorimeter is shown in **Fig. 4.1** below.

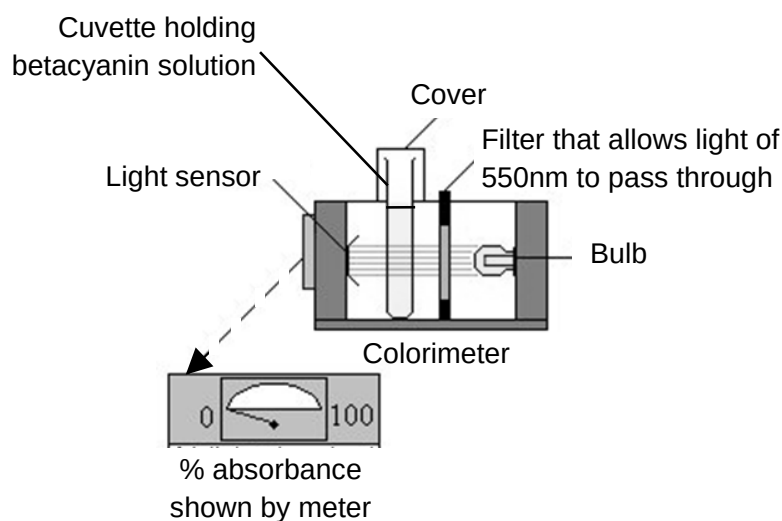


Fig. 4.1

Using the information and your own knowledge, design an experiment to investigate the effect of temperature on membrane permeability in beetroot.

Your plan must be based on the assumption that you have been provided with the following which you **must** use.

- Beet root
- Cork borer
- Scalpel
- Thermometer
- Source of boiling water and tap water
- Distilled water
- Large 500 cm³ beaker
- Colorimeter
- 2cm³ cuvettes

You may also suggest other appropriate equipment to use.

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by labeled diagrams, if necessary,
- identify independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results table(s) and graph(s) with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the entire width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the paper.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b) etc, as indicated in the question.

- 5 (a) Explain the normal functions of stem cells in a living organism. [6]
- (b) Discuss the social and ethical concerns for the use of gene therapy. [6]
- (c) Describe cystic fibrosis and how it can be treated using a non-viral gene delivery system. [8]

[Total: 20]

--- End of Paper ---